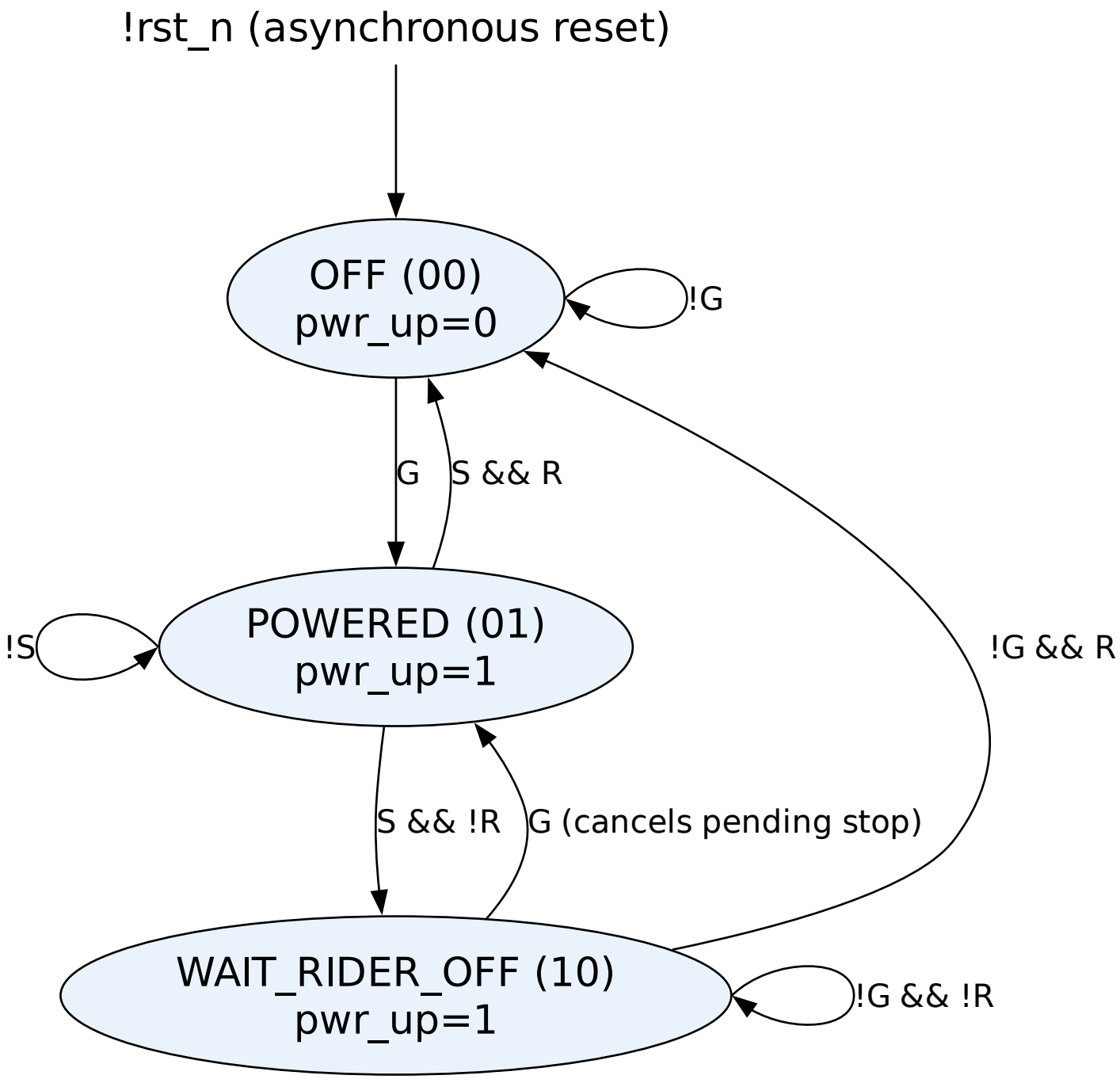
HW4 Authentication FSM - Pranav Krovi (pkrovi)

Defaults: next state = current state; pwr_up determined by state.
Every received UART byte is acknowledged: clr_rx_rdy = rx_rdy.



G = rx_rdy && rx_data == 0x47 ('G')
S = rx_rdy && rx_data == 0x53 ('S')
R = rider_off
Unrecognized bytes are ignored.
A new G has priority over R in WAIT_RIDER_OFF.
Illegal state 11: pwr_up=0, next state OFF.